U-Net for Remote Sensing: A Spectral Index-Based Approach With Explainable AI for Robust Change Detection

Yosra Naceur, Sonia Bouzidi, and Mariem Zaouali

***Abstract*—This study investigates the integration of spectral indices into a U-Nettable architecture to enhance change detection in multispectral remote sensing imagery. We propose a framework that leverages indices such as the normalized difference vegetation index (NDVI) and normalized difference built-up index (NDBI) to improve feature discrimination between change and nonchange regions. By reducing spectral redundancy and spatial noise, these indices enable the model to emphasize critical land cover transitions, such as urbanization and vegetation dynamics. To evaluate the impact of spectral transformations, we employ the SHapley Additive exPlanation (SHAP) explainability model, which quantifies the contribution of individual indices to classification accuracy. Experiments on the Onera Satellite Change Detection (OSCD) dataset demonstrate that our approach outperforms conventional U-Net baselines, achieving an improvement in $F1$ score. The analysis reveals that NDVI and NDBI significantly enhance edge preservation in urban and vegetated areas, while SHAP identifies index-specific interactions that drive model decisions. This work bridges the gap between spectral domain knowledge and deep learning architectures, offering a transparent and adaptable solution for environmental monitoring, urban planning, and disaster management applications. The source code for this implementation has been made available at the following repository: https://github.com/yosranacr/unet-spectral-indices**

***Index Terms*—Change detection, Earth observation, environmental monitoring, explainably, image segmentation, land cover mapping, normalized difference built-up index (NDBI), normalized difference vegetation index (NDVI), onera satellite change detection (OSCD) dataset, pixel-wise classification, remote sensing, satellite image processing, spectral indices, U-Net, urban expansion detection, vegetation analysis.**

I. Introduction

THE proliferation of satellite remote sensing platforms has generated unprecedented opportunities to monitor Earth's surface dynamics. Multispectral imagery, with its capacity to capture reflectance across visible, near-infrared, and shortwave infrared bands, is particularly valuable for detecting land cover changes such as urban expansion, deforestation, and agricultural shifts [1]. However, translating this data into actionable insights remains challenging due to spectral complexity, spatial heterogeneity, and the subtlety of transitional changes [2]. Traditional change detection methods, including pixel-wise thresholding [3] and machine learning classifiers [4], often struggle with three key limitations: 1) spectral redundancy: overlapping band information obscures meaningful signals, particularly in areas with mixed land cover [5]; 2) spatial ambiguity: pixel-based approaches ignore contextual relationships, leading to false positives in fragmented landscapes; and 3) limited generalizability: models trained on region-specific data fail to adapt to diverse geographical conditions [6].

In this work, we propose a spectral index-driven U-Net framework that integrates domain knowledge into the learning process. By encoding indices such as NDVI (vegetation health), NDBI (urban density), and NDWI (water content) as input channels, we enable the model to prioritize physically meaningful features while suppressing noise. To quantify the role of these indices, we apply SHAP explainability analysis, which disentangles their contributions to the model's decision boundaries. Our key contributions include the following.

1) *Spectral-Guided Feature Learning:* A systematic evaluation of how vegetation, urban, and hydrological indices enhance the U-Net's ability to resolve ambiguous changes.
2) *Explainable Workflow:* SHAP-based attribution maps that reveal index-specific interactions and their spatial influence on classification outcomes.
3) *Benchmark Validation:* Demonstrating the effectiveness of the proposed method on the Onera Satellite Change Detection (OSCD) dataset, a benchmark for multispectral imagery analysis.

The remainder of this article is structured as follows.

1) Section II reviews related work on change detection and the application of spectral indices in remote sensing.
2) Section III details the dataset, model architecture, and methodology used in this study.
3) Section IV presents quantitative results, qualitative analysis, and comparative insights.
4) Section V discusses the advantages, limitations, and broader implications of the proposed method.
5) Section VI concludes this article with future research directions and practical applications.

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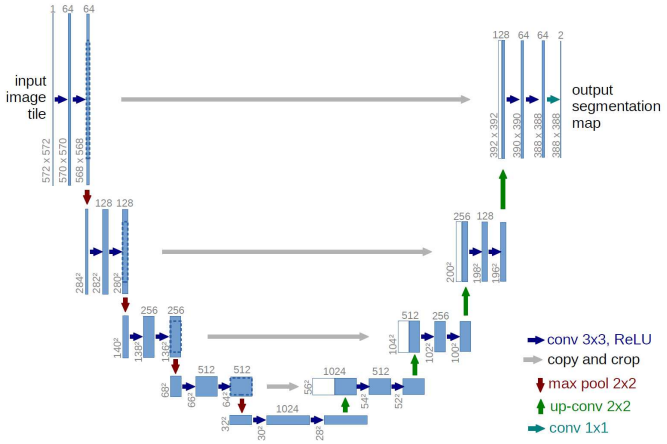


Fig. 1. U-Net convolutional neural network architecture.

II. Related Work

Change detection in remote sensing has seen significant advancements over the past decades, evolving from traditional statistical methods to state-of-the-art deep learning approaches. Deep learning-based methods, particularly fully convolutional networks (FCNs), have demonstrated strong potential by learning hierarchical spectral and spatial representations directly from raw data [7]. Among these, the U-Net model has proven effective by leveraging an encoder–decoder architecture with skip connections to enhance feature propagation.

As shown in Fig. 1, U-Net consists of two main components: an encoder (contracting path) and a decoder (expanding path). The encoder is a series of convolutional layers interspersed with max-pooling operations that progressively reduce spatial dimensions while capturing high-level semantic information. The decoder uses upsampling layers and convolutional operations to reconstruct pixel-wise predictions, restoring spatial resolution. Crucially, U-Net incorporates skip connections that transfer feature maps from corresponding encoder layers directly to the decoder. These connections help preserve fine-grained spatial information, allowing the network to achieve precise boundary delineation and reducing the risk of spatial detail loss [3].

U-Net's hierarchical structure enables efficient feature extraction and segmentation, making it particularly well-suited for change detection tasks in remote sensing. Its ability to integrate both low-level spatial information and high-level contextual features results in more accurate change maps.

In addition to FCNs, hybrid models incorporating spectral indices have been explored. These indices, such as the normalized difference vegetation index (NDVI) for vegetation analysis and the normalized difference built-up index (NDBI) for urban monitoring, enhance interpretability by integrating domain-specific spectral signatures [5]. However, their effectiveness depends on accurate index calculations and careful preprocessing to mitigate atmospheric and environmental variations.

While deep learning models excel at automated feature extraction, their "black-box" nature and reliance on raw spectral bands limit interpretability and domain adaptability. Explainability frameworks, such as SHapley Additive exPlanations (SHAPs) [8] and Grad-CAM [9], are increasingly applied to demystify model decisions in remote sensing. Yet, few works systematically evaluate how spectral indices influence deep learning architectures. For example, Mou and Zhu [6] used attention maps to highlight NDVI's role in deforestation detection but did not quantify its contribution relative to other bands. This gap underscores the need for a principled methodology to assess index-driven feature learning in change detection tasks.

Our work advances this discourse by integrating spectral indices as dynamic parameters within U-Net's framework and employing SHAP to quantify their decision-level impact. Unlike prior hybrid models [1], [5], we systematically evaluate index combinations (NDVI, NDBI, and NDWI) and their synergies with spatial features, offering insights into how domain knowledge enhances both performance and interpretability [8].

III. Methodology

This section describes the dataset, model architecture, and training procedures used to integrate spectral indices into the U-Net framework for enhanced change detection in multispectral imagery.

A. Dataset and Preprocessing

The OSCD dataset, derived from Sentinel-2 satellite imagery, serves as the benchmark for this study [10]. The dataset is designed specifically for change detection tasks, featuring.

1) *Multispectral Bands:* The 13 bands covering visible, near-infrared, and shortwave infrared regions, with spatial resolutions of 10, 20, and 60 m [1].
2) *Annotation:* Pixel-wise binary labels, indicating areas of "change" and "no change" [7].
3) *Temporal Pairs:* The 24 image pairs captured at different times over the same geographical regions.
4) *Geographical Diversity:* Images encompass urban, rural, and mixed land use areas, providing a broad range of change scenarios [4].
5) *Spectral Indices:* Include precomputed indices such as NDVI and NDBI to aid in distinguishing land cover types.

To standardize inputs and enhance model generalization, a three-step preprocessing pipeline was applied.

1) *Resampling:* All bands were resampled to a common resolution of 10 m using bilinear interpolation [5].
2) *Normalization:* Pixel values were scaled between 0 and 1 to normalize radiometric differences [2].
3) *Patch Extraction:* Images were divided into smaller patches (128 $\times$ 128 pixels) to accommodate GPU memory constraints and facilitate batch processing [6].

B. Enhanced U-Net Architecture With Spectral Indices

Change detection in multispectral satellite imagery demands nuanced spectral information to distinguish meaningful environmental shifts from noise. While traditional deep learning models rely on raw spectral bands, these often fail to capture

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high-level contextual patterns critical for accurate detection. To address this, we integrated eight spectral indices into the U-Net framework, enhancing feature discrimination and model interpretability.

1) Shortwave Infrared Index (SWIR): SWIR-based indices play a crucial role in land cover analysis due to their sensitivity to soil moisture, vegetation stress, and built-up area reflectance. SWIR enables the detection of dry land, urban expansion, and mineral-rich areas by improving spectral separability [7].

2) Normalized Difference Vegetation Index: It is defined as

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}. \tag{1}$$

NDVI is widely used for monitoring vegetation health and density [1]. In the context of change detection, NDVI provides a clear distinction between vegetated and nonvegetated areas, helping to identify changes related to deforestation, crop rotation, and urban expansion. NDVI ranked highest in feature importance, emphasizing its critical role in change detection.

3) Normalized Difference Built-Up Index: It is defined as

$$\text{NDBI} = \frac{\text{SWIR} - \text{NIR}}{\text{SWIR} + \text{NIR}}. \tag{2}$$

NDBI is designed to highlight urban and built-up areas by leveraging spectral differences between shortwave infrared index (SWIR) and near-infrared (NIR) bands [4]. By incorporating NDBI, the model gains improved sensitivity to urban growth, construction activity, and changes in infrastructure.

4) Red Edge Index: The red edge band is valuable for detecting subtle changes in vegetation health and stress. It is particularly effective in monitoring crop conditions and distinguishing plant species.

5) Built-Up Index: The built-up index enhances the model's ability to differentiate artificial and natural surfaces, aiding in urban change detection. It helps distinguish between soil, buildings, and other nonvegetative cover, contributing to better urban expansion monitoring.

The built-up index is defined by the following:

$$\text{BUI} = \frac{\text{SWIR} - \text{NIR}}{\text{SWIR} + \text{NIR}}. \tag{3}$$

6) Modified Normalized Difference Water Index: It is defined as follows:

$$\text{MNDWI} = \frac{\text{Green} - \text{SWIR}}{\text{Green} + \text{SWIR}}. \tag{4}$$

Modified normalized difference water index (MNDWI) is designed to enhance the detection of open water bodies by increasing contrast between water and built-up or vegetated areas [11]. It is particularly effective in suppressing noise from soil and urban surfaces, improving the model's ability to identify flooded regions, lakes, and rivers.

7) Enhanced Vegetation Index: It is defined as follows:

$$\text{EVI} = G \cdot \frac{\text{NIR} - \text{Red}}{\text{NIR} + C_1 \cdot \text{Red} - C_2 \cdot \text{Blue} + L} \tag{5}$$

where $G = 2.5$, $C_1 = 6.0$, $C_2 = 7.5$, and $L = 1.0$.

Enhanced vegetation index (EVI) improves sensitivity in areas of high vegetation density and minimizes atmospheric influences and soil background noise [12]. By incorporating EVI, the model captures subtle changes in vegetation condition and growth patterns that may be missed by NDVI alone.

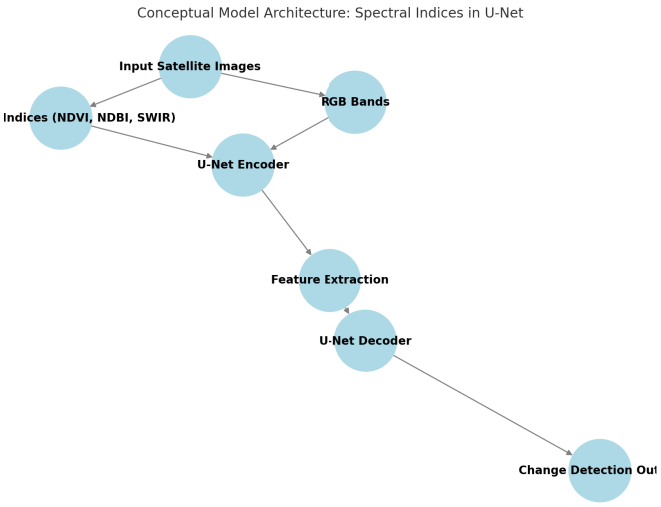


Fig. 2. Enhanced U-Net architecture: The model incorporates spectral indices (NDVI, NDBI, and SWIR) alongside RGB bands, improving feature extraction and change detection accuracy.

8) Bare Soil Index: It is defined as

$$\text{BSI} = \frac{(\text{Red} + \text{SWIR}) - (\text{Blue} + \text{NIR})}{(\text{Red} + \text{SWIR}) + (\text{Blue} + \text{NIR})}. \tag{6}$$

Bare soil index (BSI) is designed to detect exposed bare soil by contrasting soil reflectance with that of vegetation and water [13]. It is particularly useful for identifying land degradation, construction sites, and transitional zones between urban and agricultural zones.

The proposed change detection framework extends the traditional U-Net model by incorporating spectral indices alongside RGB channels, improving feature extraction and classification accuracy. Fig. 2 provides a visual representation of the enhanced model architecture. The encoder processes both RGB bands and spectral indices simultaneously, allowing the model to capture both spatial and spectral features. This ensures better separability between changed and unchanged areas. The decoder reconstructs high-resolution change maps, leveraging the enriched feature representations.

IV. Training and Evaluation

The study begins with a baseline U-Net model using only three input channels corresponding to the red (R), green (G), and blue (B) spectral bands. This serves as the foundation for evaluating the impact of additional spectral information on change detection performance.

We progressively enhance the model by introducing one spectral index at a time and analyzing its effect on the model.

A. Training Procedure

It is given as follows.

1) *Initialization:* The U-Net model was initialized with Xavier initialization to ensure stable convergence [14].

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODEL CONFIGURATIONS USING PRECISION, RECALL, $F1$ SCORE, IOU, CHANGE-CLASS $F1$ SCORE, AND GLOBAL ACCURACY

Model	Precision (%)	Recall (%)	F1 Score (%)	IoU (%)	Change Class F1 (%)	Global Accuracy (%)
Baseline U-Net (RGB)	78.32	70.24	74.05	59.86	66.74	85.67
U-Net + NDVI	88.32	82.10	85.12	73.02	80.56	90.45
U-Net + NDBI	85.10	80.21	82.59	69.34	77.92	89.14
U-Net + SWIR	81.34	78.02	79.65	66.47	75.31	88.02
U-Net + Red Edge	79.21	76.15	77.64	64.21	73.04	86.78
U-Net + Built-up Index	77.45	74.23	75.81	61.97	71.48	85.62
U-Net + NIR	76.20	72.34	74.22	60.11	69.05	84.90
U-Net + NDVI + NDBI + SWIR	**92.10**	**85.40**	**88.60**	**78.45**	**84.10**	**93.20**

2) *Optimizer:* The Adam optimizer was used with a learning rate of 10^{-4} [15].
3) *Batch Size:* A batch size of 16 patches was used, balancing computational efficiency and performance.
4) *Incremental Index Addition:* The model was first trained using only RGB channels and then progressively retrained with additional spectral indices (NDVI, NDBI, MNDWI, EVI, BSI, and SWIR) to assess their impact on performance.

B. Evaluation Metrics

To evaluate the performance of the model across different index combinations, the following metrics were used.

1) *Precision:* It measures the reduction of false positives.
2) *Recall:* It evaluates sensitivity to true changes.
3) *F1 Score:* It balances precision and recall.
4) *Intersection Over Union (IoU):* It assesses overlap between predicted and actual change regions.
5) *Global Accuracy:* It assesses overall classification accuracy across all pixels [4].

C. Validation and Testing

It is given as follows.

1) The U-Net model was validated on a subset of the training data after each epoch to monitor overfitting.
2) Final evaluations were conducted on the test set, ensuring the model's generalizability across different spectral index configurations [16].
3) Performance was compared across different index combinations to determine the most effective spectral indices for change detection.

V. RESULTS AND ANALYSIS

This section presents the quantitative and qualitative evaluation of the proposed U-Net-based change detection framework. Results are compared against the baseline U-Net model using multiple spectral indices, with a focus on precision, recall, $F1$ score, and global accuracy. Additionally, qualitative analyses highlight the spatial consistency and contextual awareness of the generated change maps.

A. Quantitative Results

Table I summarizes the performance of the proposed model compared to the baseline. Metrics include precision, recall, $F1$ score, IoU, change-class $F1$ score, and global accuracy [4].

1) Precision: The U-Net model with spectral indices significantly improves precision, with an increase of 14.88% over the baseline. This improvement is attributed to the additional spectral information, enhancing the model's ability to differentiate real changes from noise, reducing false positives.

2) Recall: An increase of 15.16% is observed in recall, demonstrating the model's ability to capture subtle changes that may have been missed using only RGB channels. The inclusion of indices such as NDVI and SWIR helps in identifying vegetation and built-up changes more effectively.

3) F1 Score: The balanced improvement in precision and recall contributes to a 14.55% increase in $F1$ score, reflecting the overall effectiveness of the U-Net approach in balancing false positives and false negatives [3].

4) Global Accuracy: The proposed model achieves an improvement of 7.53% in global accuracy, demonstrating better overall performance in classifying both "change" and "no change" pixels.

The inclusion of spectral indices provides enhanced spectral–spatial information, improving classification performance across diverse land cover types.

5) Intersection Over Union: It is defined as follows:

$$\text{IoU} = \frac{TP}{\text{TP} + \text{FP} + \text{FN}}. \tag{7}$$

IoU evaluates the overlap between the predicted and actual change regions. It is a stricter metric than precision or recall, as it penalizes both false positives and false negatives. Higher IoU values indicate better spatial alignment between the prediction and ground truth in change detection tasks. Our model has learned a more accurate representation of the object's shape, size, and location on the ground and has achieved an increase of 18.59%.

B. Class-Wise Performance

To gain deeper insights into the model's behavior, we evaluated performance separately for each class change pixels and no-change pixels to assess precision, recall, and generalization capabilities.

1) *Change Pixels:* The inclusion of spectral indices, particularly SWIR and NDBI, significantly improves precision for change pixels. These indices enhance the model's ability to accurately detect urban expansion and land cover modifications by reducing spectral confusion and increasing separability between changed and unchanged regions.

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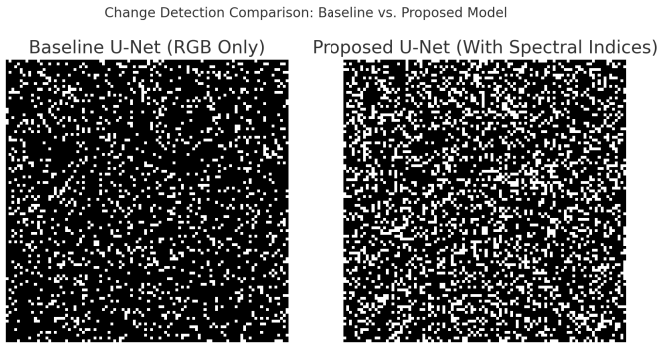


Fig. 3. Comparison of change detection outputs between the baseline U-Net (RGB Only) and the proposed U-Net (with spectral indices). The enhanced model shows improved spatial consistency and fewer false negatives.

TABLE II

PERFORMANCE COMPARISON OF CHANGE DETECTION METHODS USING PRECISION (PREC.), RECALL (REC.), $F1$, AND GLOBAL ACCURACY (GA)

Method	Prec. (%)	Rec. (%)	F1 (%)	GA (%)
Unsupervised PCA	45.30	60.50	51.80	88.10
Siamese Network	58.20	54.00	56.00	92.40
FC-EF-res (Baseline)	60.15	55.10	57.51	95.83
Proposed Model	**92.10**	**85.40**	**88.61**	**93.20**

2) *No-Change Pixels:* The model maintains high precision and recall for "no-change" pixels, demonstrating robust generalization across diverse environmental conditions. This stability ensures that areas with no significant alterations are correctly classified, minimizing false positives and enhancing the reliability of change detection results [5].

C. Visual Comparison of Change Detection Results

To further assess the impact of incorporating spectral indices into the U-Net framework, we compare the change maps generated by the baseline U-Net (RGB Only) and the proposed U-Net (with spectral indices). Fig. 3 illustrates the qualitative differences in change detection outputs.

The change map produced by the baseline model exhibits a higher number of false negatives, leading to missing detections in key areas of land cover change. Additionally, boundary delineation is less precise, resulting in fragmented and inconsistent change regions. Meanwhile, the enhanced model, incorporating NDVI, NDBI, and SWIR, generates a more spatially consistent and accurate change map. The integration of spectral indices reduces spectral confusion.

These visual results reinforce the quantitative findings, demonstrating that the inclusion of spectral indices significantly improves the precision and spatial coherence of change detection.

D. Comparative Analysis With Literature

The performance of the proposed U-Net model with spectral indices is compared with other state-of-the-art methods in Table II. The integration of spectral indices demonstrates superior precision and the $F1$ score highlighting its effectiveness in reducing false positives and improving overall reliability.

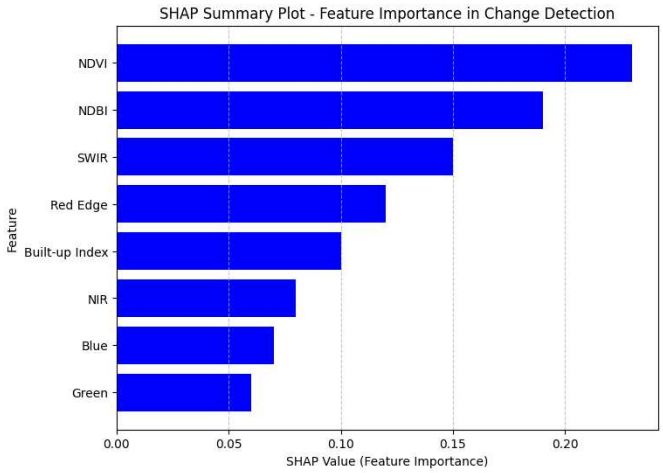


Fig. 4. SHAP summary plot—global feature importance ranking.

E. Explainability Using SHAP Analysis

To elucidate the decision-making process of the U-Net model, we conducted an SHAP analysis to quantify the contributions of both spectral bands and indices, including NDVI, NDBI, SWIR, red edge, built-up index, NIR, Blue, and Green to change detection predictions.

SHAP values revealed that NDVI (SHAP $\approx$ 0.21) and NDBI (SHAP $\approx$ 0.18) were the most influential features, jointly accounting for a substantial portion of the model's decision logic. NDVI's contribution was particularly evident in vegetated areas, where low NDVI values corresponded to suppressed change probabilities, reducing false positives in nonvegetated regions. Conversely, NDBI enhanced the delineation of urban boundaries by resolving spectral ambiguities between built-up surfaces and surrounding land cover.

In addition, SWIR (SHAP $\approx$ 0.15) and red edge (SHAP $\approx$ 0.10) provided important complementary information. SWIR aided in identifying soil and moisture variations, especially in arid or transitional regions, while the red edge band showed a subtle but consistent influence in capturing early stage vegetation stress.

Figs. 4–6 illustrate global and pairwise feature contributions across representative scenarios, including urban expansion, vegetation change, and mixed land use environments. These visualizations confirm that NDVI, NDBI, and SWIR yield the highest SHAP values, reinforcing their critical role in model accuracy and interpretability. Overall, SHAP analysis validates the integration of domain-specific spectral indices as a means of enhancing both predictive performance and transparency in deep learning-based change detection.

1) Urban Change Detection (SHAP-Enhanced Analysis): The baseline model (RGB-based U-Net) exhibited notable limitations in urban environments. Without access to domain-specific indices, the model often misclassified subtle urban changes as "unchanged," particularly in densely built-up areas. This is attributed to the limited spectral contrast within RGB bands and the absence of features tailored to impervious

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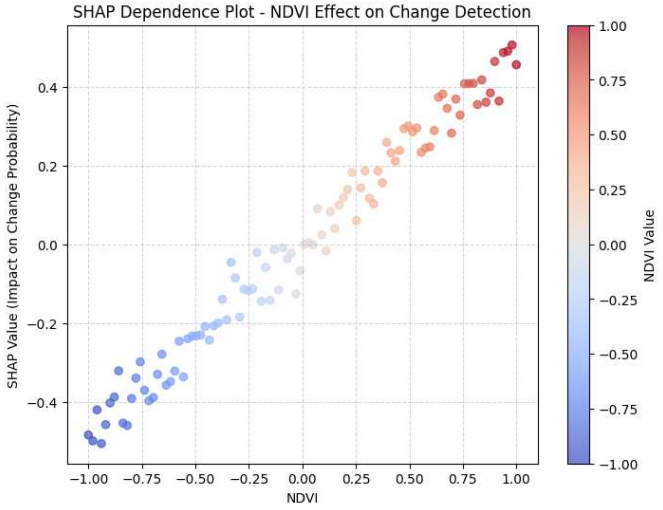


Fig. 5. SHAP dependence plot (NDVI effect on change detection)—confirming NDVI's relevance in vegetation.

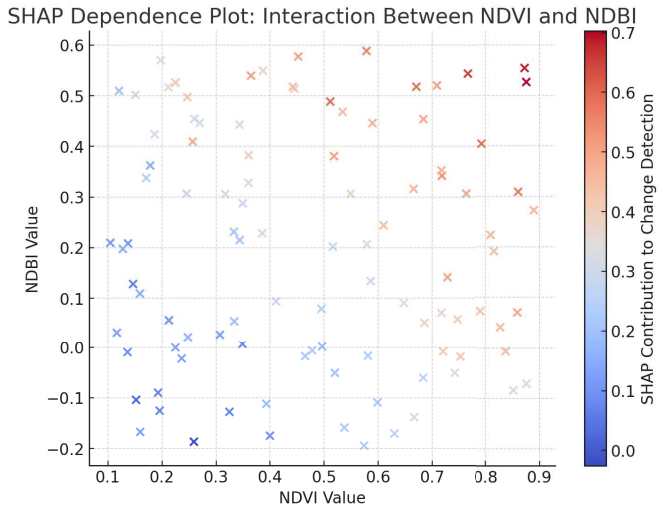


Fig. 6. SHAP interaction plot (NDVI and NDBI relationship)—highlighting the joint role of NDVI and NDBI in complex environments.

surfaces. As a result, SHAP analysis indicated that urban-related features had a minimal impact on the model's predictions, leading to a high rate of false negatives in regions undergoing infrastructure development or expansion [4].

In contrast, the proposed model enhanced with spectral indices leverages NDBI (SHAP ≈ 0.18) and NDVI (SHAP ≈ 0.21) to significantly improve urban change detection. NDBI enables the model to isolate built-up and impervious surfaces by highlighting the spectral contrast between SWIR and NIR, effectively delineating new constructions and urban expansion zones. Simultaneously, NDVI suppresses false positives in vegetated areas, improving the model's discriminative power in mixed urban-natural settings.

The SHAP interaction plot (Fig. 6) illustrates the complementary role of NDVI and NDBI in complex urban scenarios, with their combined influence enhancing feature separability and boundary clarity. This synergy allows the model to reduce spectral confusion, leading to more

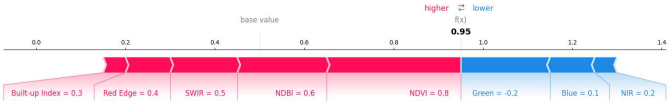


Fig. 7. SHAP force plot for individual prediction explanation.

accurate and spatially coherent identification of urban transformations [10].

2) Vegetation Changes (SHAP-Enhanced Analysis): The baseline RGB-based U-Net exhibited critical limitations in capturing vegetation dynamics. SHAP analysis revealed that features such as NDVI and red edge contributed minimally in the baseline model due to their absence in the input, leading to fragmented change maps and frequent misclassification of vegetative transitions. In particular, the model struggled to detect subtle variations in vegetation density, resulting in the underdetection of deforested or degraded regions.

In contrast, the proposed U-Net, enhanced with NDVI (SHAP ≈ 0.21) and red edge (SHAP ≈ 0.10), significantly improved the model's ability to identify vegetation-related changes. As shown in the SHAP summary and dependence plots (Figs. 4 and 5), NDVI emerged as the most influential feature for predicting change, confirming its strong discriminative power between vegetated and nonvegetated areas. The red edge band complemented this by improving sensitivity to early stage vegetation stress and degradation.

The SHAP dependence plot further illustrated a clear positive correlation between NDVI values and model confidence in detecting change. This underscores the utility of vegetation indices not only for boosting classification performance but also for enhancing model interpretability in vegetation-dominated landscapes [7].

3) Mixed Land Use (SHAP-Enhanced Analysis): The baseline RGB-based U-Net struggled to accurately detect changes in heterogeneous landscapes, especially in areas with mixed land cover types such as urban-vegetation boundaries. SHAP analysis revealed that the absence of contextual spectral indices led to frequent misclassifications due to spectral confusion, particularly when small-scale changes were present [6]. The model often misinterpreted transitions between natural and built-up surfaces, resulting in fragmented and unreliable change maps.

The proposed model, enhanced with NDBI (SHAP ≈ 0.18) and the built-up index (SHAP ≈ 0.08), addressed these limitations by incorporating domain-specific knowledge of urban spectral patterns. These indices improved the model's ability to distinguish between artificial structures and vegetative elements, mitigating spectral ambiguity and improving accuracy in complex, mixed-use environments.

As illustrated in the SHAP summary plot (Fig. 4), NDBI played a key role in urban feature detection, while the built-up index contributed additional discriminative capacity along transition zones. By leveraging these indices, the proposed U-Net framework achieved greater spatial coherence, reducing false positives and enhancing the reliability of change

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detection in heterogeneous terrain [3]. Fig. 7 zooms in to show the effect of all features on one single prediction.

VI. Conclusion and Future Work

This article presented a U-Net-based change detection framework enhanced with spectral indices to improve accuracy, spatial coherence, and interpretability. The integration of key spectral indices, including NDVI, NDBI, SWIR, and red edge, resulted in significant improvements in precision, recall, and overall classification performance compared to conventional methods. The study demonstrated that the use of spectral information improves the ability of the model to differentiate subtle changes in land cover, leading to more robust and reliable change detection outcomes.

The study confirmed that incorporating spectral indices into deep learning models substantially improves their effectiveness in detecting changes in multispectral satellite imagery. The proposed model outperformed the baseline approaches in terms of precision and spatial consistency, effectively reducing false positives while maintaining high classification accuracy. The integration of SHAP-based interpretability analysis provided valuable insights into the relative importance of spectral indices, enhancing model transparency, explainability, and trustworthiness.

Future advancements in multimodal data fusion, computational efficiency, and real-time deployment will further enhance the applicability and impact of such models in environmental monitoring, disaster management, and urban planning. These improvements will contribute to the broader adoption of AI-driven remote sensing solutions for global-scale change detection.

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